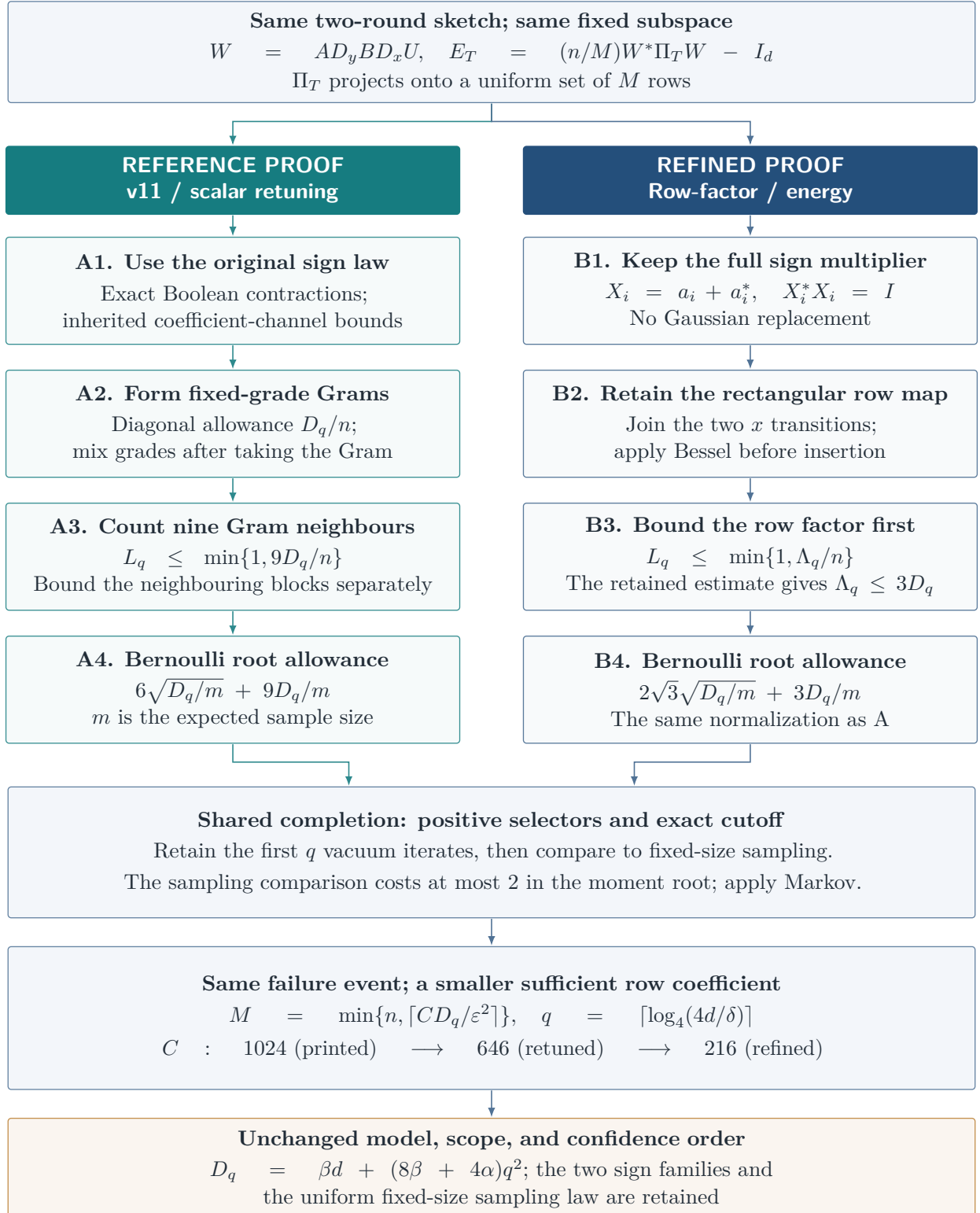


Two-round SRHT

Where the row-factor refinement improves the constant

Read downward. Teal shows the reference proof; blue shows the refinement.

Full-width boxes identify the common model, completion, and scope.



Proof and constants comparison

Same two-round law; a sharper row-factor estimate.

Both routes use the same coherence allowance D_q , moment order q , and uniform fixed-size sampling law. The table separates a stronger operator inequality from merely retuning the final probability certificate.

Comparison	Reference proof Supplied v11	Refined proof Row-factor / energy (§4)
Retained object	Fixed-grade row Grams, assembled by their nine possible neighbours.	The rectangular row factor; keep both x transitions joined during y creation.
Local coefficient budget	D_q/n on each input grade.	The same inherited channel estimates; no new local law.
Subset-effect allowance	$L_q \leq \min\{1, 9D_q/n\}$.	$L_q \leq \min\{1, \Lambda_q/n\}$, with $\Lambda_q \leq 3D_q$.
Bernoulli root allowance	$6\sqrt{D_q/m} + 9D_q/m$	$2\sqrt{3}\sqrt{D_q/m} + 3D_q/m$
Fixed-size root allowance	$12\sqrt{D_q/M} + 18D_q/M$	$4\sqrt{3}\sqrt{D_q/M} + 6D_q/M$
Row coefficient	1024 as printed; 646 after scalar retuning alone.	216, using the retained row-factor estimate.
Unchanged scope	Arbitrary fixed isometry and coherent complex unitary pairs.	Same model and D_q ; the same $4q$ -wise sign requirement, capped at n .

Root allowances exclude the common external factor $d^{1/(2q)}$. Here $m = \rho n$ for Bernoulli sampling and M is the fixed sample size.

Sufficient row counts

Flat transforms; $\varepsilon = 1/2$, $\delta = 0.01$, $n = 2^{30}$. All columns use $q = \lceil \log_4(400d) \rceil$ and $D_q = d + 12q^2$. The full comparison appears in Section 6.

d	q	Printed: 1024	Retuned: 646	Refined: 216
10	6	1 810 432	1 142 128	381 888
100	8	3 555 328	2 242 912	749 952
1 000	10	9 011 200	5 684 800	1 900 800
10 000	11	46 907 392	29 591 968	9 894 528

These are sufficient theorem prescriptions, not empirical thresholds. The middle column prevents numerical retuning from being credited to the new contraction.

Quantitative endpoint

$$M = \min\{n, \lceil 216D_q/\varepsilon^2 \rceil\} \implies \mathbb{P}\{\|E_T\| > \varepsilon\} \leq \delta.$$

The improvement $1024 \rightarrow 646$ is retuning; $646 \rightarrow 216$ uses the stronger contraction. For flat transforms at confidence 0.99, the rank-only coefficient is 65016 instead of 308224.

Universal improvement

For the same two-round sketch, the same arbitrary fixed subspace, and the same coherence allowance

$$D_q = \beta d + (8\beta + 4\alpha)q^2,$$

we reduce the subset-effect loss from $9D_q/n$ to $3D_q/n$. The structural step bounds the *rectangular row map before its Gram*: the two first-layer transitions remain joined while the second layer creates an occupation.

The resulting fixed-size sampling theorem permits

$$M = \min\{n, \lceil 216\varepsilon^{-2}D_q \rceil\}$$

with the same $q = \lceil \log_4(4d/\delta) \rceil$ as v11, instead of its coefficient 1024. A control calculation shows that retuning the old proof alone gives 646; thus the contraction improvement is separate from numerical retuning.

These are analytic refinements of the supplied appendices. The Rademacher signs and sampling law are unchanged. Gaussian Wick pairings without the original-law corrections would not justify the proof.

1 The precise theorem and the baseline

Let $A, B \in \mathbb{C}^{n \times n}$ be unitary and $U \in \mathbb{C}^{n \times d}$ satisfy $U^*U = I_d$. Set

$$\alpha = n \max_{j,a} |A_{ja}|^2, \quad \beta = n \max_{a,i} |B_{ai}|^2. \quad (1.1)$$

Let D_x, D_y be independent diagonal matrices of independent unbiased signs, and put

$$W = AD_yBD_xU, \quad w_j^* = e_j^*W.$$

For a uniform M -element subset $T \subseteq [n]$, independent of the signs, let R_T restrict coordinates to T . The sketch and error are

$$S = \sqrt{n/M} R_T AD_yBD_x, \quad E_T = U^* S^* S U - I_d. \quad (1.2)$$

For Walsh matrices, this is the two-round SRHT model; the theorem also covers arbitrary complex unitary pairs with their stated coherences. The term “SRHT” below refers to this two-round model, not to the classical single-round construction studied in [3].

Theorem 1.1 (Improved fixed-size sampling constant). *Let $0 < \varepsilon < 1$ and $0 < \delta < 1/2$. Define*

$$\begin{aligned} q &= \lceil \log_4(4d/\delta) \rceil, & D_q &= \beta d + (8\beta + 4\alpha)q^2, \\ M &= \min\{n, \lceil 216D_q/\varepsilon^2 \rceil\}. \end{aligned} \quad (1.3)$$

Then every fixed isometry U satisfies

$$\mathbb{P}\{\|E_T\| > \varepsilon\} \leq \delta. \quad (1.4)$$

Each sign family may instead be $\min(n, 4q)$ -wise independent, provided the two families and the uniform subset remain mutually independent.

The supplied v11 theorem has exactly the same D_q , q , and sampling model, with 1024 instead of 216 [1, coherent-unitary extension and coefficient-kernel appendix]. We retain its stronger quadratic-order allowance rather than comparing against the earlier cubic-order statement. The proof below inherits the coefficient contractions from v11 and replaces the *assembly of their norm bounds*. The local double-annihilation reflection estimate is not claimed as new.

2 Where Wick must be adapted to the sign law

For unbiased independent signs,

$$\mathbb{E}[\epsilon_i \epsilon_j \epsilon_k \epsilon_l] = \delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk} - 2\mathbf{1}_{\{i=j=k=l\}}. \quad (2.1)$$

The first three terms are Gaussian pairings; the final term is the sign fourth-cumulant correction. It is indispensable: a single sign has fourth moment one rather than three. Isserlis' theorem [2] is the Gaussian pairing identity, not a formula for the sign law without this correction.

Use the exact Boolean occupation basis indexed by coordinate subsets. The annihilator a_i removes i when present, and a_i^* inserts it when absent. Put $N_i = a_i^* a_i$ and $X_i = a_i + a_i^*$. On the *full* occupation space,

$$X_i^* X_i = I, \quad a_i a_i^* = a_i^* a_i + \delta_{il}(I - 2N_i). \quad (2.2)$$

The factor $I - 2N_i$ is the original-law occupation correction. At distinct sites the operators commute. The corresponding identities in the second sign family use y instead of x .

The v11 appendix already uses (2.2) to group a positive coefficient kernel with its reflection. The present refinement keeps an additional exact relation, $X_i^* X_i = I$, through the row-factor contraction instead of splitting both x transitions and then reconstructing a Gram by a nine-neighbor estimate. This is the valid adaptation of the rMPS hybrid [5]: exact local contractions, including non-Gaussian corrections, followed by a retained energy bound. It is not a change of probability space or a Gaussian comparison theorem.

3 The inherited coefficient estimates

Let $\mathcal{X}_r, \mathcal{Y}_s$ be the r - and s -subset occupation spaces. The inner product is conjugate linear in its first argument. Set

$$c_i = e_i^* U, \quad (g_{ji})_a = \sqrt{n} \overline{A_{ja} B_{ai}}, \quad a_y(g) = \sum_a \overline{g_a} a_{y,a}. \quad (3.1)$$

For complex coefficients, the full second-layer multiplier is

$$Y(g) = a_y(g) + a_y(\bar{g})^* = \sum_a \overline{g_a} (a_{y,a} + a_{y,a}^*).$$

The row multiplication map is

$$R_j = \frac{1}{\sqrt{n}} \sum_i c_i \otimes X_i \otimes Y(g_{ji}). \quad (3.2)$$

For a set T , collect these rows as $R_T f = (R_j f)_{j \in T}$, retaining the row label as an orthogonal output register. The channel $R_T^{\mu\nu}$ changes the (x, y) grades by $(\mu, \nu) \in \{+1, -1\}^2$.

3.1 Bessel relations and insertion costs

Unitarity and (1.1) imply

$$\begin{aligned} \|\sum_i z_i g_{ji}\|^2 &\leq \alpha \sum_i |z_i|^2 && \text{for each } j, \\ \sum_j |g_{ji}\rangle \langle g_{ji}| &\preceq \beta I && \text{for each } i, \quad \sum_i c_i^* c_i = I_d. \end{aligned} \quad (3.3)$$

For example the first synthesis matrix is $\sqrt{n} \operatorname{diag}(\overline{A_{j,:}}) \overline{B}$, whose norm is at most $\sqrt{\alpha}$. The second is $\sqrt{n} \operatorname{diag}(\overline{B_{:,i}}) A^*$, whose norm is at most $\sqrt{\beta}$.

Extracting one coordinate from grade s has norm $\sqrt{s}$: the map sends $|J\rangle$ to $\sum_{a \in J} e_a \otimes |J \setminus \{a\}\rangle$, whose squared norm is s . Its adjoint inserts a coordinate and has norm $\sqrt{s+1}$ on inputs of grade s . Hilbert-valued coefficient extensions of (3.3) follow by expansion in an orthonormal basis of the passive space.

Lemma 3.1 (Input-grade contraction bounds from v11). *On fixed input grades (r, s) , with $t = |T|$, put $p = (r-1)_+$ and $v = (s-1)_+$. Then*

$$n[(R_T^{++})^* R_T^{++} + (R_T^{-+})^* R_T^{-+}] \preceq \alpha t(s+1)I, \quad (3.4)$$

$$n\|R_T^{+-}\|^2 \leq \beta(r+1)s, \quad (3.5)$$

$$n\|R_T^{--}\|^2 \leq \beta(d+p+pv) + \alpha tv. \quad (3.6)$$

Double annihilation is zero if $r = 0$ or $s = 0$; the nonnegative right side of (3.6) is still a permissible bound there.

Proof inputs and attribution. These are the joint-creation, mixed-channel, and collective double-annihilation lemmas of [1, coefficient-map section and coefficient-kernel appendix]. Their exact roles are as follows.

For (3.4), form the collected coefficient map $\sum_i \overline{g_{ji}} \otimes (c_i \otimes X_i)$. Equation (3.3) and $X_i^* X_i = I$ bound it by $\sqrt{\alpha}$, before an insertion of norm $\sqrt{s+1}$. At fixed r its two x outputs are orthogonal, giving the sum of Grams. We prove and use its stronger mixed- x -grade form in Section 4.

For (3.5), extract one y coordinate, apply the collected map

$$(F_T z)_{j,i} = (c_i \otimes \langle g_{ji}|) z, \quad F_T^* F_T = \sum_i c_i^* c_i \otimes \sum_{j \in T} |g_{ji}\rangle \langle g_{ji}| \preceq \beta I,$$

and insert one x coordinate. The costs multiply to $\sqrt{\beta(r+1)s}$.

For (3.6), the output Gram on grades (p, v) has blocks

$$\mathcal{G}_{jk} = \sum_{i,l} P_{il} a_{x,i} a_{x,l}^* \otimes a_y(g_{ji}) a_y(g_{kl})^*, \quad P = U U^*.$$

Its exact Boolean normal ordering gives five terms $\mathcal{E} + \mathcal{X} - \mathcal{F} + \mathcal{Y} + \mathcal{Z}$. The four *grouped* bounds proved in the appendix are

$$\|\mathcal{E}\| \leq \beta d, \quad \|\mathcal{X} - \mathcal{F}\| \leq \beta p, \quad \|\mathcal{Y}\| \leq \alpha tv, \quad \|\mathcal{Z}\| \leq \beta pv.$$

The improved second term retains $\mathcal{X} - \mathcal{F}$ as a direct sum of positive coefficient lifts times $I - 2N_{y,a}$; it is not obtained by dropping a negative term. The last term is an extracted two-species coefficient kernel of norm at most β , lifted at cost pv . These precisely specified kernel estimates are inherited, not reasserted as a new Wick lemma. Their sum proves (3.6). $\square$

The fixed-input diagonal Gram budget obtained by adding these bounds is

$$n\|\Pi_{r,s}R_T^*R_T\Pi_{r,s}\| \leq \beta[d + p + pv + (r + 1)s] + \alpha t(v + s + 1). \quad (3.7)$$

For $r, s \leq 2q$ and $t \leq q$, it is at most D_q . The old proof extends (3.7) to all mixed input grades using positivity and nine Gram neighbors. This gives $9D_q/n$. We next replace that extension.

4 Retain the row factor: the loss is three, not nine

Let P_{sig} retain both input sign grades at most $\min(2q, n)$. *Do not truncate the output of R_T when forming its squared norm.* Then

$$P_{\text{sig}}M_{\sum_{j \in T} w_j w_j^*}P_{\text{sig}} = (R_T P_{\text{sig}})^*(R_T P_{\text{sig}}).$$

This equality is what permits use of the full-space identity $X_i^* X_i = I$ in a finite-cutoff proof.

Define

$$\begin{aligned} a_q &= \alpha q(2q + 1), \\ b_q &= \beta(d + 8q^2) + \alpha q(2q - 1), \\ \Lambda_q &= (\sqrt{a_q} + \sqrt{2b_q})^2. \end{aligned} \quad (4.1)$$

Notice $a_q + b_q = D_q$ and $\Lambda_q \leq 3D_q$ by Cauchy–Schwarz.

Theorem 4.1 (Hybrid subset-effect estimate). *For every $q \geq 1$ and $T \subseteq [n]$ with $|T| \leq q$,*

$$\boxed{\|R_T P_{\text{sig}}\|^2 \leq \Lambda_q/n \leq 3D_q/n.} \quad (4.2)$$

In particular the positive row effects

$$A_j^{\text{eff}} = P_{\text{sig}}M_{w_j w_j^*}P_{\text{sig}}$$

sum to the identity on the retained sign space, and their subset allowance satisfies

$$\boxed{L_q := \max_{|T| \leq \min(q, n)} \left\| \sum_{j \in T} A_j^{\text{eff}} \right\| \leq \min\{1, \Lambda_q/n\} \leq \min\{1, 3D_q/n\}.} \quad (4.3)$$

Proof. Split the row map by the direction of its y transition:

$$R_T = \mathcal{C}_T + \mathcal{A}_T, \quad \mathcal{C}_T = R_T^{++} + R_T^{-+}, \quad \mathcal{A}_T = R_T^{+-} + R_T^{--}.$$

Here each sum denotes the actual operator across all input grades, not a sum of separate worst-case norms.

Creation retains both x transitions.

Fix a y grade s but allow arbitrary mixed x grades. Before the final y insertion, the row- j coefficient map is

$$G_j f = \sum_i \bar{g}_{ji} \otimes (c_i \otimes X_i \otimes I) f.$$

Bessel synthesis, valid with passive Hilbert factors, gives

$$\|G_j f\|^2 \leq \alpha \sum_i \|(c_i \otimes X_i \otimes I)f\|^2 = \alpha \|f\|^2.$$

The equality uses the full x multiplier X_i , not either of its partial transitions. Insert the new y coordinate at cost $\sqrt{s+1}$ and collect t rows. Hence on this entire mixed- x space,

$$n\|\mathcal{C}_T f\|^2 \leq \alpha t(s+1)\|f\|^2.$$

Distinct input y grades have distinct output y grades under creation, so their ranges are orthogonal. Consequently

$$\|\mathcal{C}_T P_{\text{sig}}\|^2 \leq \alpha q(2q+1)/n = a_q/n. \quad (4.4)$$

Annihilation has only two incoming grades at each output.

Decompose $f = \sum_{r,s} f_{r,s}$ by its input grades. A fixed output pair (r', s') under $\mathcal{A}_T$ can receive contributions only from $(r' - 1, s' + 1)$ and $(r' + 1, s' + 1)$. Cauchy–Schwarz on these two vectors, followed by summing the orthogonal output grades, yields

$$\|\mathcal{A}_T f\|^2 \leq 2 \sum_{r,s} \|\mathcal{A}_T f_{r,s}\|^2.$$

For a fixed input pair, the two x transitions have orthogonal outputs. Lemma 3.1 therefore gives

$$n\|\mathcal{A}_T f_{r,s}\|^2 \leq \{\beta[d + p + pv + (r+1)s] + \alpha tv\} \|f_{r,s}\|^2.$$

Over $r, s \leq 2q$ and $t \leq q$, the braces are bounded by b_q : at the endpoint,

$$p + pv + (r+1)s = 8q^2, \quad tv \leq q(2q-1).$$

Thus

$$\|\mathcal{A}_T P_{\text{sig}}\|^2 \leq 2b_q/n. \quad (4.5)$$

Combine only after the two retained estimates.

The triangle inequality applied to (4.4) and (4.5) proves (4.2). The effects are positive, and their sum is the identity because $W^*W = I_d$ pointwise. Their subset sums are therefore also bounded by I , proving (4.3). $\square$

4.1 Exactly where the improvement enters

The old argument bounded the Gram by nine neighbors in the (x, y) grade lattice. Bounding the row factor first already gives a generic improvement to $4D_q/n$, because each row-map output receives at most four input grade pairs. Theorem 4.1 improves 4 to 3 by *not splitting the x grades at all in the y -creation part*. The identity $X_i^* X_i = I$ then retains the relevant cancellation before any cross-grade Cauchy–Schwarz step.

The diagonal allowance D_q itself is unchanged. Nor is every map with a nine-neighbor Gram guaranteed the factor three; the stronger result uses the exact two-layer factorization and the retained full x multiplier.

5 Complete the moment proof on the original selector law

We include the remaining operator argument to make the constants independently checkable. The ingredients are the positive-effects and finite-sampling lemmas already in v11 [1].

5.1 Positive effects and selector creation

For arbitrary positive operators A_j with $\sum_j A_j = I$, define

$$L_q = \max_{|T| \leq q} \left\| \sum_{j \in T} A_j \right\|.$$

On the selector subset space of grades at most q , set

$$(Kf)_T = \sum_{j \in T} A_j f_{T \setminus \{j\}}, \quad (Df)_J = \left(\sum_{j \in J} A_j \right) f_J.$$

Then

$$\|K\| \leq \sqrt{L_q}, \quad \|D\| \leq L_q. \quad (5.1)$$

Indeed, for each output set T , factor the positive row through $(A_j^{1/2})_{j \in T}$ to obtain

$$\left\| \sum_{j \in T} A_j f_j \right\|^2 \leq \left\| \sum_{j \in T} A_j \right\| \sum_{j \in T} \langle f_j, A_j f_j \rangle.$$

Sum over sets T of one fixed cardinality. Each input J is charged $\sum_{j \notin J} A_j \preceq I$, giving the creation bound. Different input grades have orthogonal creation ranges. The occupancy bound is block diagonal.

5.2 Bernoulli moment bound

For independent selectors $\eta_j \sim \text{Bernoulli}(\rho)$, independent of the signs, put

$$E_\rho = \rho^{-1} W^* \text{diag}(\eta) W - I_d, \quad m = \rho n.$$

In the normalized basis $1, (\eta - \rho)/\sqrt{\rho(1-\rho)}$, multiplication by $\eta/\rho - 1$ is

$$\begin{pmatrix} 0 & \sigma \\ \sigma & \tau \end{pmatrix}, \quad \sigma = \sqrt{(1-\rho)/\rho}, \quad \tau = (1-2\rho)/\rho.$$

Apply (5.1) to the effects of Theorem 4.1. Compress sign grades to $2q$ and selector grade to q . The multiplier is

$$M_{\text{cut}} = \sigma(K + K^*) + \tau D,$$

so

$$\|M_{\text{cut}}\| \leq 2\sigma\sqrt{L_q} + |\tau|L_q \leq 2\sqrt{\Lambda_q/m} + \Lambda_q/m. \quad (5.2)$$

One multiplication raises each sign grade by at most two and selector grade by at most one. Thus the first q iterates from the vacuum agree exactly with original-law multiplication, and

$$\begin{aligned} \mathbb{E} \text{tr} E_\rho^{2q} &= \sum_{a=1}^d \|M_{\text{cut}}^q(e_a \otimes \Omega)\|^2 \\ &\leq d[2\sqrt{\Lambda_q/m} + \Lambda_q/m]^{2q} \\ &\leq d[2\sqrt{3}\sqrt{D_q/m} + 3D_q/m]^{2q}. \end{aligned} \quad (5.3)$$

The dimension factor is d , not the dimension of the cutoff space. Keeping the finite sampling density in (5.2) also permits the sharper root

$$2\sqrt{(1-\rho)\Lambda_q/m} + |1-2\rho|\Lambda_q/m,$$

but no use of that additional refinement is made in the constant 216.

5.3 Fixed-size sampling and its exact barycenter

Let $1 \leq M < n$, $\rho = M/n$, and let $H_1, \dots, H_n$ be deterministic Hermitian matrices. For a Bernoulli subset $\mathcal{B}$ and a uniform M -subset $\mathcal{T}$, define

$$Z_{\mathcal{B}} = \sum_j (\mathbf{1}_{j \in \mathcal{B}} - \rho) H_j, \quad Z_{\mathcal{T}} = \sum_j (\mathbf{1}_{j \in \mathcal{T}} - \rho) H_j.$$

There is a coupling with

$$\mathbb{E}[Z_{\mathcal{B}} \mid \mathcal{T}] = c Z_{\mathcal{T}}, \quad c = 1 - \mathbb{P}\{\text{Bin}(n-1, \rho) = M\} \geq \frac{1}{2}. \quad (5.4)$$

To construct it, draw $\mathcal{B}$ with size K and remove uniform elements if $K > M$, or add uniform missing elements if $K < M$. Symmetry makes $\mathcal{T}$ uniform independently of K . With

$$a = \mathbb{E}(K - M)_+ = \mathbb{E}(M - K)_+, \quad v = n\rho(1 - \rho),$$

the conditional inclusion probabilities inside and outside $\mathcal{T}$ are $1 - a/M$ and $a/(n - M)$. Their difference is $c = 1 - a/v$.

Let $Y \sim \text{Bin}(n-1, \rho)$. The binomial identity $k \binom{n}{k} = n \binom{n-1}{k-1}$ gives

$$a = M[\mathbb{P}(Y \geq M) - \mathbb{P}(Y \geq M+1)] = M(1 - \rho)\mathbb{P}(Y = M) = v\mathbb{P}(Y = M).$$

The neighboring atoms $\mathbb{P}(Y = M)$ and $\mathbb{P}(Y = M-1)$ are equal, because their ratio is $\frac{n-M}{M} \frac{\rho}{1-\rho} = 1$. Therefore each is at most $1/2$, proving (5.4).

The function $Z \mapsto \text{tr } Z^{2q} = \|Z\|_{S_{2q}}^{2q}$ is convex on Hermitian matrices. Conditional Jensen yields

$$\mathbb{E} \text{tr } Z_{\mathcal{T}}^{2q} \leq c^{-2q} \mathbb{E} \text{tr } Z_{\mathcal{B}}^{2q} \leq 2^{2q} \mathbb{E} \text{tr } Z_{\mathcal{B}}^{2q}. \quad (5.5)$$

This is the centered Bernoulli-to-fixed-size coupling of v11. The classical sampling literature, including [4], concerns related comparisons; we do not substitute a without-replacement versus with-replacement statement for (5.5).

Apply (5.5) conditionally on the signs to $H_j = w_j w_j^*$ and rescale by ρ^{-1} . With (5.3) this proves

$$\boxed{\mathbb{E} \text{tr } E_T^{2q} \leq d[4\sqrt{\Lambda_q/M} + 2\Lambda_q/M]^{2q} \leq d[4\sqrt{3}\sqrt{D_q/M} + 6D_q/M]^{2q}.} \quad (5.6)$$

For $M = n$ the error is zero by unitarity; no transfer is needed.

5.4 Proof of the sampling constant and moment transfer

If $M \geq 216D_q/\varepsilon^2$, the last root in (5.6) is at most

$$\varepsilon \left(\frac{\sqrt{2}}{3} + \frac{\varepsilon}{36} \right) \leq \varepsilon \frac{12\sqrt{2} + 1}{36} < \frac{\varepsilon}{2}. \quad (5.7)$$

The last strict inequality is the exact integer check $288 < 289$. Markov now gives $\mathbb{P}\{\|E\|_T > \varepsilon\} < d4^{-q} \leq \delta/4 < \delta$, proving Theorem 1.1.

Each entry of E_T has degree at most two in each sign family. A word in $\text{tr } E_T^{2q}$ therefore involves at most $4q$ distinct coordinates of either sign family. Matching those moments transfers the full-independent proof to the stated limited-independent law. Similarly, the Bernoulli result allows $\min(n, 2q)$ -wise independent selectors with correct marginals, independent of both sign families. The fixed-size sample retains its uniform law; it is not assumed to have independent coordinates.

6 Constants: structural improvement versus retuning

The two changes should not be conflated. The row-factor estimate improves the *proof inequality*; selecting 216 then makes that inequality imply the desired failure bound. Some improvement to the printed 1024 is possible even without the new estimate.

Quantity	Supplied v11	Retained row-factor proof
Fixed-input diagonal budget	D_q/n	unchanged
Subset-effect allowance	$9D_q/n$	$\Lambda_q/n \leq 3D_q/n$
Bernoulli root	$6\sqrt{D_q/m} + 9D_q/m$	$2\sqrt{3}\sqrt{D_q/m} + 3D_q/m$
Fixed-size root	$12\sqrt{D_q/M} + 18D_q/M$	$4\sqrt{3}\sqrt{D_q/M} + 6D_q/M$
Sampling prescription coefficient	1024 printed; 646 after retuning	216
Sign independence requirement	$\min(n, 4q)$ per family	unchanged
Flat-transform allowance	$d + 12q^2$	unchanged

Table 1: The same model, isometry, moment order, and coherence parameters. A subset allowance is also capped by 1. The table omits this cap only to display the constants.

Retuning control.

With the old root, a coefficient 646 already suffices:

$$\frac{12}{\sqrt{646}} + \frac{18}{646} < \frac{1}{2}, \quad 576 \cdot 646 = 372096 < 372100 = 610^2.$$

Thus $1024 \rightarrow 646$ is a scalar retuning of the existing theorem, not a hybrid contraction gain. The comparison $646 \rightarrow 216$ isolates the benefit at this conservative half-error target, approximately a 66.6% reduction. Relative to the printed 1024, the reduction is 78.9%. Neither 646 nor 216 is asserted optimal for all possible proof refinements or all choices of q .

6.1 Concrete row prescriptions

For flat A, B , let $\varepsilon = 1/2$, $\delta = 0.01$, and take $n = 2^{30}$ so none of the displayed prescriptions is capped at n . All columns use the same $q = \lceil \log_4(400d) \rceil$.

d	q	D_q	v11: 1024	Old retuned: 646	New: 216
10	6	442	1 810 432	1 142 128	381 888
100	8	868	3 555 328	2 242 912	749 952
1 000	10	2 200	9 011 200	5 684 800	1 900 800
10 000	11	11 452	46 907 392	29 591 968	9 894 528

Table 2: Sufficient theorem row counts, not empirical sample requirements. The middle column prevents scalar retuning from being attributed to the new contraction.

6.2 The fixed-confidence rank-only prescription

The v11 estimate $D_q = d + 12q^2 \leq 301d$ at $q = \lceil \log_4(400d) \rceil$ gives the following immediate corollary:

Corollary 6.1 (Flat two-round transforms, confidence 0.99). *For $0 < \varepsilon < 1$, flat unitary A, B , and each fixed rank- d isometry, taking*

$$M = \min\{n, \lceil 65016d/\varepsilon^2 \rceil\} \quad (6.1)$$

is sufficient for failure probability at most 0.01. The original printed coefficient is 308224.

For clarity, the elementary rank estimate can be checked without a numerical scan. For $q = 5$, $q^2/d \leq 25$. If $q = k \geq 6$, then $d > 4^{k-1}/400$, so $q^2/d < 400k^2/4^{k-1} \leq 400 \cdot 36/4^5 < 25$; the latter sequence decreases for $k \geq 6$. Thus $1 + 12q^2/d \leq 301$, and $216 \cdot 301 = 65016$.

7 Verification, attribution, and limits

The script `hybrid_appendices_checks.py` verifies the integer comparisons in (5.7) and the 646 retuning, recomputes all entries of Table 2, and checks small real and complex row operators built directly from Boolean masks. The latter tests compare creation, annihilation, and full row norms against (4.4), (4.5), and (4.2). They are supplementary numerical checks, not substitutes for the analytic inequalities. No Monte Carlo evidence is used.

The new theorem does not reduce the quadratic confidence exponent, improve the coefficient 12 inside $d + 12q^2$, change the number of transform rounds, or alter the uniform fixed-size sampling law. The original confidence obstruction and the all-row endpoint remain relevant. The more precise Λ_q may improve individual certificates further, but Table 2 consistently uses the simple universal envelope $3D_q$.

The original kernel and selector lemmas are attributed to the supplied version, including its already improved reflection grouping. The added step is the whole-row bound before Gram relaxation. This is a genuine strengthening of that proof's inequality, but not evidence that a Wick-only, Boolean-chaos, or other operator argument could not derive it independently. The proof is analytic, not an external review or a Lean certificate; the formalization claims associated with earlier statements must not be transferred to this refinement.

References

- [1] Diar Heidary. *Graded Multiplication Operators for Random-Matrix Moments*. Version 11, exposition and attribution revision, 16 September 2026. Supplied source `v11_attribution_revised.tex`. Used here: the coherent-unitary theorem; collective row-channel bounds; exact Boolean double-annihilation kernels; positive-effects lemma; centered finite-sampling coupling; finite vacuum compression. The new proof retains these inputs and improves their cross-grade assembly.
- [2] Leon Isserlis. On a formula for the product-moment coefficient of any order of a normal frequency distribution in any number of variables. *Biometrika* 12(1–2), 134–139, 1918. <https://doi.org/10.1093/biomet/12.1-2.134>. Gaussian pairing identity; the Boolean corrections here are established from the original sign law.
- [3] Joel A. Tropp. Improved analysis of the subsampled randomized Hadamard transform. *Advances in Adaptive Data Analysis* 3(1–2), 115–126, 2011. <https://arxiv.org/abs/1011.1595>. Classical single-round SRHT background, not the two-round constant theorem used as the baseline here.
- [4] David Gross and Vincent Nesme. *Note on sampling without replacing from a finite collection of matrices*. arXiv:1001.2738, 2010. <https://arxiv.org/abs/1001.2738>. Related finite-sampling comparisons; the specific centered Bernoulli barycenter in Section 5.3 is the stated lemma from v11 and is proved here.
- [5] Method Problems working note. *Gaussian rMPS Moments: Two Proof Chains and a Wick–Energy Hybrid*. 16 September 2026. Supplied source `rmgs_three_chains_hybrid.tex`. Motivation for exact contraction data followed by a continuation-valid energy estimate; no Gaussian-core theorem is applied to signs without its required corrections.